

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0346

Snohomish County Hilton Lake 230 kV
April 27, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0346. The Interconnection Customer proposes a 135.6 MW wind facility interconnecting at the Snohomish County Hilton Lake 230 kV substation in the Illinois area. The request was received on June 20, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 168 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$4,609,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Brevard - Snohomish County Hilton Lake 230 KV	Carlyss - Minonk Wind Farm 500 KV	107.9	1,019	91.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brevard - Snohomish County Hilton Lake 230 KV	Reconductor line	\$6,730,000	68% (\$4,609,000)
	Total allocated to project		\$4,609,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.